

Unit 1 - AM

MC - Questions Answered
on D2L

MC	Written	Total
/18	/20	/38

Knowledge/Application

- Given a polynomial function with constant fifth differences of -1440. State the following:
 - Name of Function _____
 - Leading Coefficient _____
- Describe the transformation that must be applied to the graph of the function $f(x) = x^3$ to obtain the transformed function $y = -3f\left(\frac{1}{2}x - 4\right) - 2$.
 - Sketch the transformed function using mapping rule and showing at least 5 distinct points.

Application

- Determine algebraically whether $f(x) = (2x - 2)(x + 4)(3x - 12)(x + 1)$ is an even function, odd function or neither.
- Sketch the graph for the following function; $f(x) = .0052083(2x - 1)(x + 4)^3(x - 3)^2(x + 1)$, make sure to label all key points on your graph.

Communication

- Discuss the relationship between the degree of a polynomial function and the following features of the corresponding graph:
 - the number of x-intercepts
 - the number of maximum and minimum points
- Is it possible for a polynomial function with constant positive fourth differences to have no zeros and a y-intercept of (0,-4)? If not, explain.

Thinking

- Given the following table, determine the equation of the function

x	y
1	2
2	0
3	-3
4	-4
5	0
6	12